

Exercise 1A Solutions Key

Comprehensive Study Guide & Solutions Manual

Grounded in the Course Syllabus and Textbook Notes

Theoretical Core: Definition of Integers

As established in the textbook, the set of all integers, denoted by $\mathbb{I}$ or $\mathbb{Z}$, is formed by all counting numbers $(1, 2, 3, \dots)$, zero (0) , and the negatives of counting numbers $(-1, -2, -3, \dots)$:

$$\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$$

- $\mathbb{I}^+ = \{1, 2, 3, \dots\} = \mathbb{N}$ is the set of all positive integers (counting numbers).
- $\mathbb{I}^- = \{-1, -2, -3, \dots\}$ is the set of all negative integers.
- The number 0 is an integer that is neither positive nor negative.

Question 1: Finding Integers Between Ranges

Instruction: Write all integers between the given numbers.

Key Principle: Interval Exclusivity

When finding integers “between” a and b (where $a < b$), we are looking for the set of integers x such that:

$$a < x < b$$

This interval is strictly open, meaning the boundary values a and b are **excluded** from the solution.

(i) **Between -9 and -3 :**

We look for integers x such that $-9 < x < -3$. Starting from the integer immediately to the right of -9 and moving rightward on the number line:

Solution: $-8, -7, -6, -5, -4$

(ii) **Between -33 and -26 :**

We look for integers x such that $-33 < x < -26$.

Solution: $-32, -31, -30, -29, -28, -27$

(iii) **Between -5 and 2 :**

We look for integers x such that $-5 < x < 2$. This interval spans across negative integers, zero, and positive integers.

Solution: $-4, -3, -2, -1, 0, 1$

(iv) **Between -1 and 6 :**

We look for integers x such that $-1 < x < 6$.

Solution: $0, 1, 2, 3, 4, 5$

Question 2: Comparison of Integers

Instruction: Compare the integers and insert the correct inequality symbol ($<$ or $>$).

Key Principle: Number Line Ordering

On a standard horizontal number line, any integer situated to the right of another integer is greater.

- Every positive integer is greater than 0, and 0 is greater than every negative integer.
- For any two positive integers a and b , if $a < b$, then $-a > -b$ (the inequality reverses when negated).

(i) **Compare -16 and -61 :**

On the number line, -16 lies to the right of -61 because its numerical distance from zero is smaller. Alternatively, since $16 < 61$, the negation reverses the inequality:

$$-16 > -61$$

Answer: $-16 > -61$

(ii) **Compare -39 and 13 :**

Every negative integer lies to the left of zero, and every positive integer lies to the right. Therefore, a negative integer is always less than a positive integer:

$$-39 < 13$$

Answer: $-39 < 13$

(iii) **Compare -236 and -362 :**

Since $236 < 362$, negating both values reverses the inequality sign:

$$-236 > -362$$

Answer: $-236 > -362$

(iv) **Compare -2 and 0 :**

Zero is always situated to the right of any negative integer on the number line. Therefore, every negative integer is less than zero:

$$-2 < 0$$

Answer: $-2 < 0$

Question 3: Absolute Value & Modulus Evaluation

Instruction: Evaluate the mathematical expressions.

Key Principle: Absolute Value / Modulus

The absolute value of an integer a , denoted by $|a|$, represents its numerical distance from zero on the number line, regardless of its sign. It is defined as:

$$|a| = \begin{cases} a, & \text{if } a \geq 0 \\ -a, & \text{if } a < 0 \end{cases}$$

Thus, the modulus of a negative integer is always its positive counterpart. For example, $|-11| = 11$ and $|23| = 23$.

- (i) **Evaluate** $20 - |-11|$:

First, find the absolute value: $|-11| = 11$. Then substitute it back into the expression:

$$20 - 11 = 9$$

Solution: 9

- (ii) **Evaluate** $|-23| - |-16|$:

Evaluate each modulus: $|-23| = 23$ and $|-16| = 16$. Substitute and solve:

$$23 - 16 = 7$$

Solution: 7

- (iii) **Evaluate** $|-137| + |13|$:

Evaluate each modulus: $|-137| = 137$ and $|13| = 13$. Substitute and solve:

$$137 + 13 = 150$$

Solution: 150

- (iv) **Evaluate** $6 - |-4|$:

First, find the absolute value: $|-4| = 4$. Substitute and solve:

$$6 - 4 = 2$$

Solution: 2

Question 4: Arranging in Ascending Order

Instruction: Arrange the given sets of integers in ascending order (from smallest to largest).

Key Principle: Ascending Ordering Strategy

To arrange a mixed set of integers in ascending order:

1. Group and sort the negative integers (larger negative magnitude is smaller).
2. Position zero (0) if present.
3. Group and sort the positive integers (larger magnitude is larger).

(i) **Set:** $\{-9, 11, -23, 41, -37, 0, -5\}$

- Negative integers sorted: $-37 < -23 < -9 < -5$

- Zero: 0

- Positive integers sorted: $11 < 41$

Combining these groups gives:

Solution: $-37, -23, -9, -5, 0, 11, 41$

(ii) **Set:** $\{-31, 19, -50, -8, -23, 3, 8\}$

- Negative integers sorted: $-50 < -31 < -23 < -8$

- Positive integers sorted: $3 < 8 < 19$

Combining these groups gives:

Solution: $-50, -31, -23, -8, 3, 8, 19$

(iii) **Set:** $\{-2, 12, -43, 31, 7, -35, -10\}$

- Negative integers sorted: $-43 < -35 < -10 < -2$

- Positive integers sorted: $7 < 12 < 31$

Combining these groups gives:

Solution: $-43, -35, -10, -2, 7, 12, 31$

Question 5: Arranging in Descending Order

Instruction: Arrange the given sets of integers in descending order (from largest to smallest).

Key Principle: Descending Ordering Strategy

To arrange a mixed set of integers in descending order:

1. Group and sort positive integers from largest to smallest.
2. Position zero (0) if present.
3. Group and sort negative integers from largest (closest to zero) to smallest (farthest from zero).

(i) **Set:** $\{-24, 16, -40, -5, -13, 6, -1\}$

- Positive integers sorted: $16 > 6$

- Negative integers sorted: $-1 > -5 > -13 > -24 > -40$

Combining these groups gives:

Solution: $16, 6, -1, -5, -13, -24, -40$

(ii) **Set:** $\{0, -7, 19, -23, -3, 8, 46\}$

- Positive integers sorted: $46 > 19 > 8$

- Zero: 0

- Negative integers sorted: $-3 > -7 > -23$

Combining these groups gives:

Solution: $46, 19, 8, 0, -3, -7, -23$

(iii) **Set:** $\{-31, -13, -49, 4, 37, -9, -52\}$

- Positive integers sorted: $37 > 4$

- Negative integers sorted: $-9 > -13 > -31 > -49 > -52$

Combining these groups gives:

Solution: $37, 4, -9, -13, -31, -49, -52$

Question 6: Theoretical Foundation (Fill in the Blanks)

Instruction: Complete the statements with the correct mathematical terms.

- (i) 0 is greater than every negative integer.

Explanation: Since zero lies to the right of every negative integer on the number line, it is mathematically greater than all of them.

- (ii) Every negative integer is less than every positive integer.

Explanation: All negative integers lie to the left of zero, whereas all positive integers lie to the right of zero, meaning negative numbers are always strictly smaller.

- (iii) For any two positive integers a and b , if $a < b$, then $-a > \underline{\text{(greater than)}} -b$.

Explanation: Multiplying an inequality by a negative number reverses the direction of the inequality sign.

- (iv) Modulus of a negative integer is always positive.

Explanation: By definition, absolute value (or modulus) represents a distance from zero, which must be non-negative. For any non-zero negative integer x , $|x| > 0$.

- (v) The largest negative integer is -1.

Explanation: As we move to the right along the negative portion of the number line towards zero, the values increase. The rightmost negative integer immediately preceding zero is -1 .

- (vi) The smallest positive integer is 1.

Explanation: The positive integers begin at 1 and increment by 1 indefinitely ($\mathbb{I}^+ = \{1, 2, 3, \dots\}$), making 1 the smallest value in the set.